

ELEC2134: Class Test 2 (Fourier Transform)

Date: Thursday 5 March 2026 (Week 3)

Time: 3 - 4 pm (Exam Duration 40 minutes).

Instruction to candidates: To maximise your marks, show all steps clearly when deriving your answers (do not write final answers only).

Pass Level Questions. Total Marks: 64

Question 1:

Part A: Consider the rectangular pulse defined as

$$x(t) = \begin{cases} 1 & -2 \leq t \leq 2 \\ 0 & |t| > 2 \end{cases}$$

(a) Find the Fourier Transform $X(\omega)$ of $x(t)$. [20 Marks]

(b) Find the maximum value of $X(\omega)$. [5 Marks]

(c) Find the value of ω at which $X(\omega)$ first becomes zero. [5 Marks]

(d) Sketch the magnitude spectrum $|X(\omega)|$. [10 Marks]

(e) Sketch the phase spectrum $\angle X(\omega)$. [10 Marks]

Question 1:

Part B: The signal $x(t)$ from Part A above is multiplied by $\cos(10t)$ to form the modulated signal

$$y(t) = x(t)\cos(10t).$$

Sketch the magnitude spectrum $|Y(\omega)|$ clearly indicating the frequency locations and amplitudes.

[14 Marks]

Higher Level Questions. Total Marks: 36

Question 2: Figure 1 shows a triangular pulse $y(t)$:

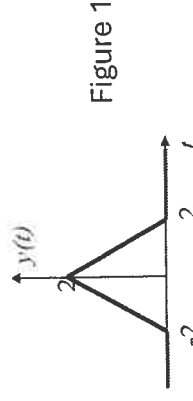


Figure 1

Using the time-differentiation property, show the Fourier transform $Y(\omega)$ of the triangular pulse shown in Figure 1 is given by :

$$Y(\omega) = \frac{4}{\omega^2} \sin^2 \omega$$

[36 Marks]

Hint: Use the time differentiation property twice.

$$\frac{d}{dt} x(t) \xleftrightarrow{FT} j\omega X(\omega); \quad \frac{d^n}{dt^n} x(t) \xleftrightarrow{FT} (j\omega)^n X(\omega)$$

Note: The derivative of a signal at a jump discontinuity produces an impulse at that discontinuity, with amplitude equal to the size of the jump.

Common Signals and Their Fourier Transforms

$f(t)$	$F(\omega)$
$\delta(t)$	1
A	$2\pi A \delta(\omega)$
$\text{sgn}(t)$	$\frac{2}{j\omega}$
$u(t)$	$\pi \delta(\omega) + \frac{1}{j\omega}$
$e^{-at}u(t), a > 0$	$\frac{1}{a + j\omega}$
$e^{at}u(-t), a > 0$	$\frac{1}{a - j\omega}$
$e^{-at} t , a > 0$	$\frac{2a}{\omega^2 + a^2}$
$e^{j\omega_0 t}$	$2\pi \delta(\omega - \omega_0)$
$\cos(\omega_0 t)$	$\pi [\delta(\omega - \omega_0) + \delta(\omega + \omega_0)]$
$\sin(\omega_0 t)$	$j\pi [\delta(\omega + \omega_0) - \delta(\omega - \omega_0)]$

Fourier Transform Properties

Function of Time	Fourier Transform
$f(t)$	$F(\omega)$
$Af(t)$	$AF(\omega)$
$a f_1(t) + b f_2(t)$	$a F_1(\omega) + b F_2(\omega)$
$f(t - t_0)$	$e^{-j\omega t_0} F(\omega)$
$f(at), a > 0$	$\frac{1}{a} F\left(\frac{\omega}{a}\right)$
$e^{j\omega_0 t} f(t)$	$F(\omega - \omega_0)$
$\frac{d^n f(t)}{dt^n}$	$(j\omega)^n F(\omega)$
$\int_{-\infty}^{\infty} f_1(\tau) f_2(t - \tau) d\tau$	$F_1(\omega) F_2(\omega)$
$t^n f(t)$	$(j)^n \frac{d^n F(\omega)}{d\omega^n}$
$f(-t)$	$F(-\omega)$
$\int_{-\infty}^t f(\tau) d\tau$	$\frac{F(\omega)}{j\omega} + \pi F(0) \delta(\omega)$

$$\begin{aligned}\sin(\alpha \pm \beta) &= \sin \alpha \cos \beta \pm \cos \alpha \sin \beta \\ \cos(\alpha \pm \beta) &= \cos \alpha \cos \beta \mp \sin \alpha \sin \beta \\ \cos(\alpha \pm 90^\circ) &= \mp \sin \alpha\end{aligned}$$

$$\sin(\alpha \pm 90^\circ) = \pm \cos \alpha$$

$$\cos \alpha \cos \beta = \frac{1}{2} \cos(\alpha + \beta) + \frac{1}{2} \cos(\alpha - \beta)$$

$$\sin \alpha \sin \beta = \frac{1}{2} \cos(\alpha - \beta) - \frac{1}{2} \cos(\alpha + \beta)$$

$$\sin \alpha \cos \beta = \frac{1}{2} \sin(\alpha + \beta) + \frac{1}{2} \sin(\alpha - \beta)$$

$$\sin 2\alpha = 2 \sin \alpha \cos \alpha$$

$$\cos 2\alpha = 2 \cos^2 \alpha - 1 = 1 - 2 \sin^2 \alpha = \cos^2 \alpha - \sin^2 \alpha$$

$$\sin^2 \alpha = \frac{1}{2} (1 - \cos 2\alpha)$$

$$\cos^2 \alpha = \frac{1}{2} (1 + \cos 2\alpha)$$

$$\sin \alpha = \frac{e^{j\alpha} - e^{-j\alpha}}{2j}$$

$$\cos \alpha = \frac{e^{j\alpha} + e^{-j\alpha}}{2}$$

$$e^{\pm j\alpha} = \cos \alpha \pm j \sin \alpha$$

$$A \cos \alpha + B \sin \alpha = \sqrt{A^2 + B^2} \cos \left(\alpha + \tan^{-1} \left(\frac{B}{A} \right) \right)$$

1. Integrals (for integer $n \neq 0$)

$$\begin{aligned}\int \sin(ax) dx &= -\frac{1}{a} \cos(ax); & \int \cos(ax) dx &= \frac{1}{a} \sin(ax) \\ \int_0^T \sin(n\omega_0 t) dt &= 0; & \int_0^T \cos(n\omega_0 t) dt &= 0\end{aligned}$$

2. Integration by Parts (Key Formula)

$$\int u dv = uv - \int v du$$

$$\int x e^{ax} dx = \frac{e^{ax}}{a^2} (ax - 1); \quad \int x^2 e^{ax} dx = \frac{e^{ax}}{a^3} (a^2 x^2 - 2ax + 2)$$

5. Definite Integration by Parts: For $0 \leq t \leq T$ and integer n :

$$\int_0^T t \sin(n\omega_0 t) dt = \left[\frac{-T}{n\omega_0} \right] \quad \text{and} \quad \int_0^T t \cos(n\omega_0 t) dt = \left[\frac{0}{n\omega_0} \right]$$

3. Integration by Parts: (a) $\int t \cos(n\omega_0 t) dt$

$$\text{Let } u = t \Rightarrow du = dt; \quad dv = \cos(n\omega_0 t) dt \Rightarrow v = \frac{1}{n\omega_0} \sin(n\omega_0 t)$$

$$\int t \cos(n\omega_0 t) dt = \frac{t}{n\omega_0} \sin(n\omega_0 t) + \frac{1}{(n\omega_0)^2} \cos(n\omega_0 t) + \text{Constant}$$

4. Integration by Parts: (b) $\int t \sin(n\omega_0 t) dt$

$$\text{Let } u = t \Rightarrow du = dt; \quad dv = \sin(n\omega_0 t) dt \Rightarrow v = -\frac{1}{n\omega_0} \cos(n\omega_0 t)$$

$$\int t \sin(n\omega_0 t) dt = -\frac{t}{n\omega_0} \cos(n\omega_0 t) + \frac{1}{(n\omega_0)^2} \sin(n\omega_0 t) + \text{Constant}$$

Fourier Transform pair:

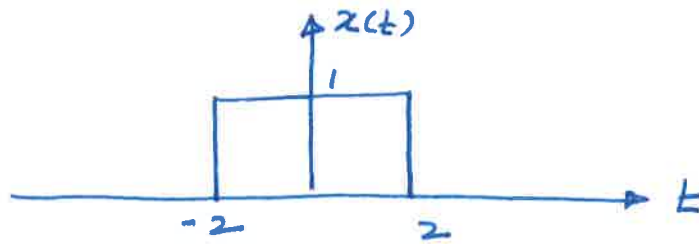
$$X(\omega) = \int_{-\infty}^{\infty} x(t) e^{-j\omega t} dt \quad (\text{FT})$$

$$x(t) = \frac{1}{2\pi} \int_{-\infty}^{\infty} X(\omega) e^{j\omega t} d\omega \quad (\text{IFT})$$

$$\text{Modulation: } FT\{x(t) \cos(\omega_0 t)\} = \frac{1}{2} X(\omega - \omega_0) + \frac{1}{2} X(\omega + \omega_0)$$

$$\int_{-\infty}^{+\infty} e^{-x^2/2\sigma^2} dx = \sigma\sqrt{2\pi} \quad \sigma > 0 \quad \int_{-\infty}^{\infty} e^{-a u^2} du = \sqrt{\frac{\pi}{a}}, \quad a > 0$$

Q₁



(a) $X(\omega) = \int_{-\infty}^{\infty} x(t) e^{-j\omega t} dt$

$$= \int_{-2}^2 1 e^{-j\omega t} dt = \left[\frac{e^{-j\omega t}}{-j\omega} \right]_{-2}^2$$

$$= \frac{e^{-j\omega 2} - e^{-j\omega(-2)}}{-j\omega} = \frac{2}{\omega} \frac{e^{j2\omega} - e^{-j2\omega}}{2j}$$

$$= \frac{2}{\omega} \sin 2\omega$$

20 Marks

(b) $X(\omega) = 4 \left(\frac{\sin 2\omega}{2\omega} \right)$ $\frac{\sin 2\omega}{2\omega} \rightarrow 1$ when $\omega \rightarrow 0$

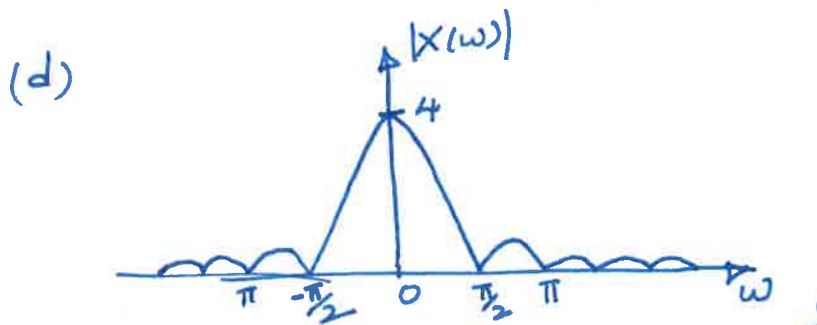
$\therefore X(\omega)|_{\text{max}} = 4$

5 Marks

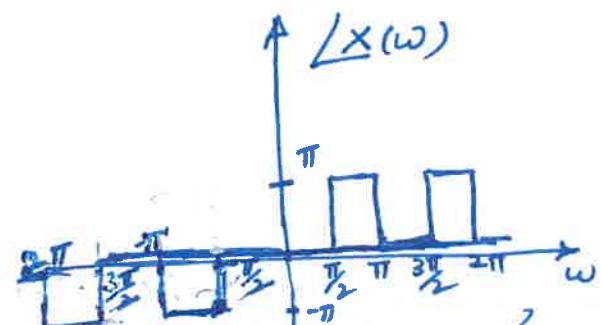
(c) $X(\omega) = \frac{2}{\omega} \sin 2\omega$

$\sin 2\omega = 0 \Rightarrow \boxed{\omega = \pm \pi/2}$

5 Marks.

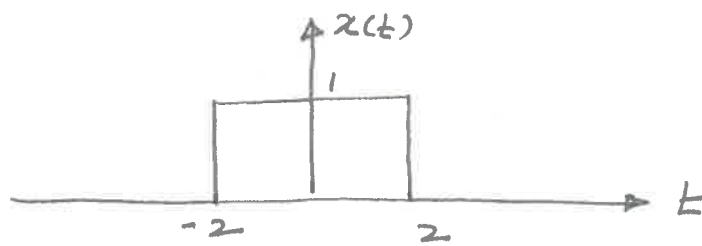


10 Marks



10 Marks.

Q₁



(a)
$$X(\omega) = \int_{-\infty}^{\infty} x(t) e^{-j\omega t} dt$$

$$= \int_{-2}^2 1 e^{-j\omega t} dt = \left[\frac{e^{-j\omega t}}{-j\omega} \right]_{-2}^2$$

$$= \frac{e^{-j\omega 2} - e^{-j\omega(-2)}}{-j\omega} = \frac{2}{\omega} \frac{e^{j2\omega} - e^{-j2\omega}}{2j}$$

$$= \frac{2}{\omega} \sin 2\omega$$

20 Marks

(b)
$$X(\omega) = 4 \frac{\sin 2\omega}{2\omega}$$

$$\frac{\sin 2\omega}{2\omega} \rightarrow 1 \text{ when } \omega \rightarrow 0$$

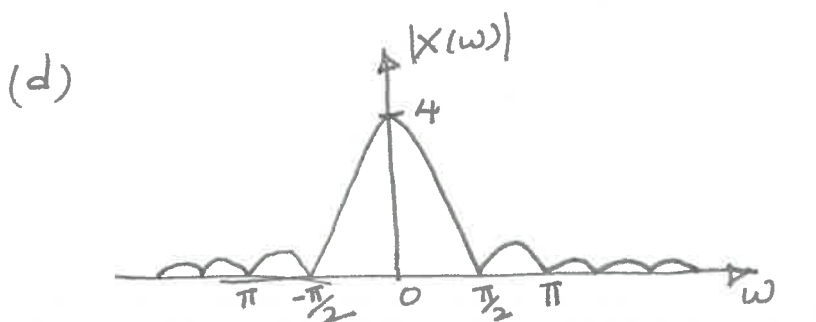
$$\therefore X(\omega) \Big|_{\text{max}} = 4$$

5 Marks

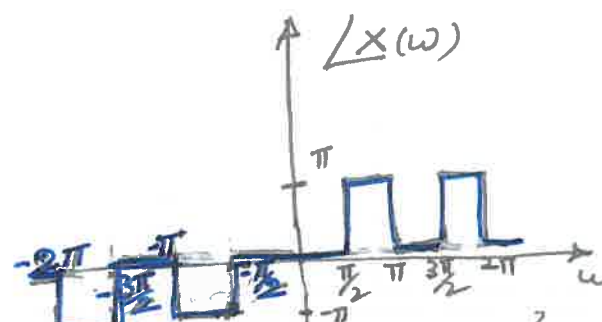
(c)
$$X(\omega) = \frac{2}{\omega} \sin 2\omega$$

$$\sin 2\omega = 0 \Rightarrow \omega = \pm \frac{\pi}{2}$$

5 Marks.



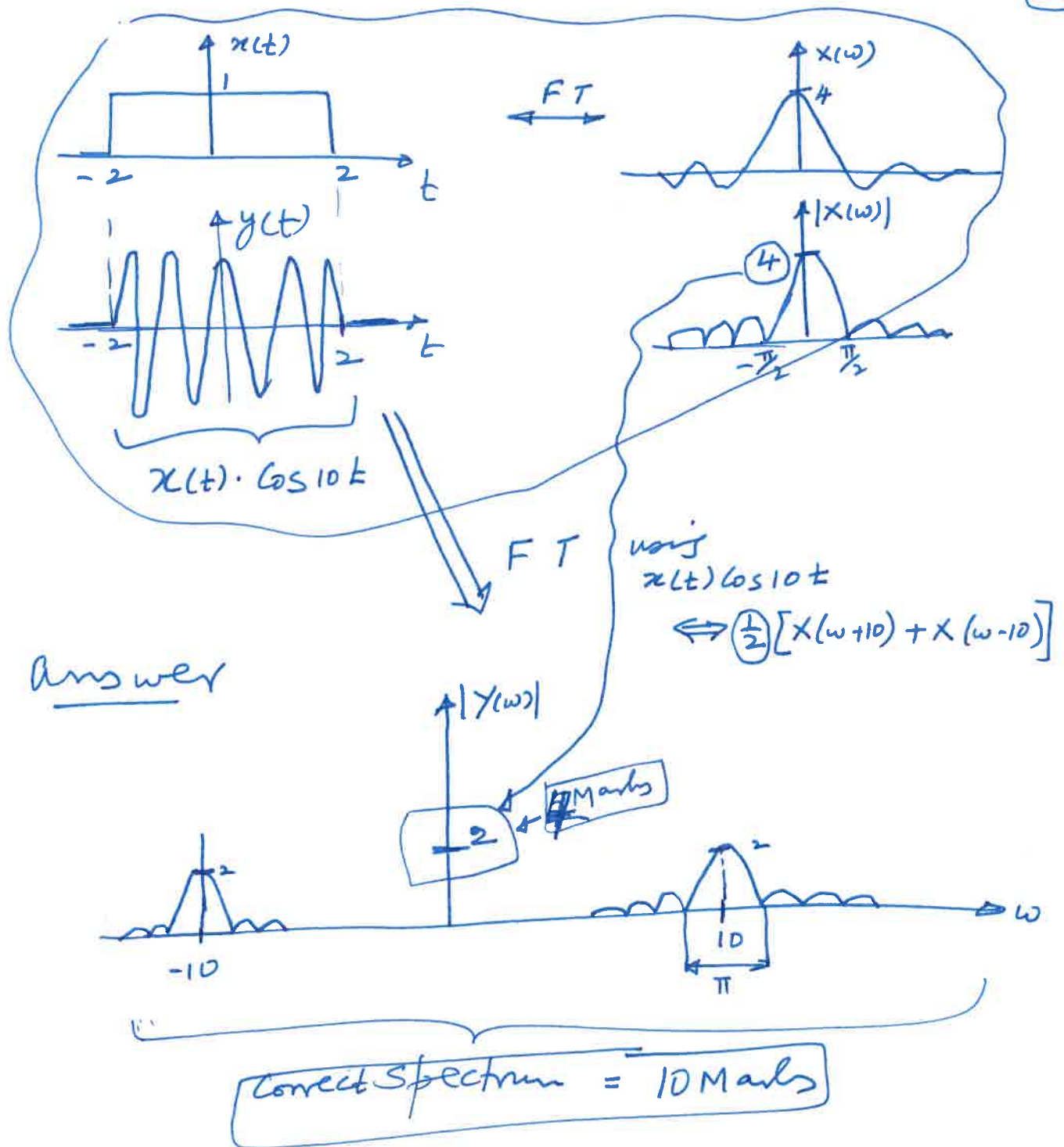
10 Marks



10 Marks.

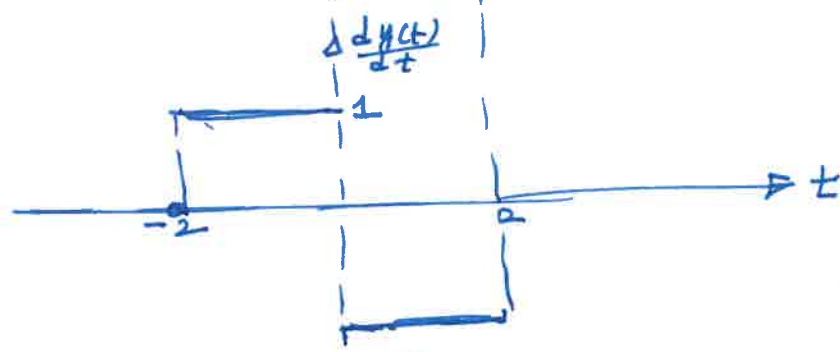
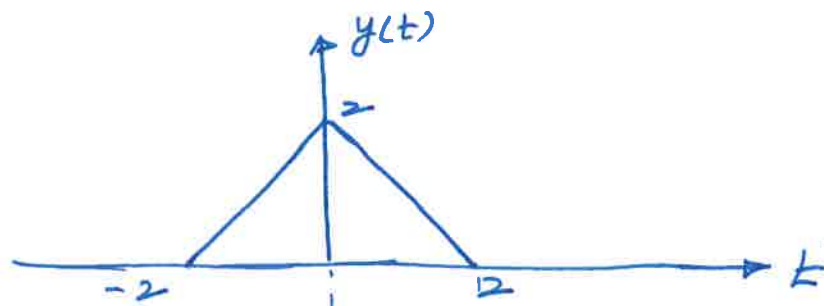
Q1
(b)

2

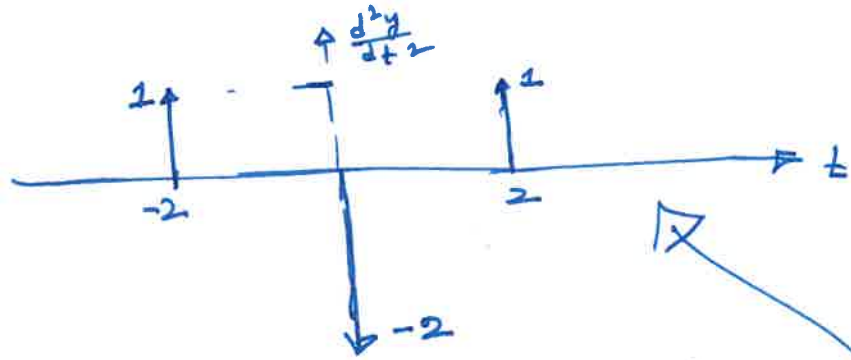


Question 2

(3)



5 Marks



5 Marks

using time differentiation

5 Marks $\rightarrow \frac{d^2 y(t)}{dt^2} \Leftrightarrow (j\omega)^2 Y(\omega) = -\omega^2 Y(\omega)$ — (1)

5 Marks using time-shifting Property $\delta(t-t_0) \Leftrightarrow e^{-j\omega t_0}$ — (2)

Use (1) & (2)

16 Marks

$$\begin{aligned}
 -\omega^2 Y(\omega) &= 1 \cdot e^{-j\omega 2} + (-1) e^{-j\omega \cdot 0} + 1 \cdot e^{j\omega 2} \\
 -\omega^2 Y(\omega) &= \frac{e^{j2\omega} + e^{-j2\omega}}{2} - 1 \\
 &= \cos 2\omega - 1 \\
 &= 2(1 - \sin^2 \omega) - 2 \\
 -\omega^2 Y(\omega) &= -4 \sin^2 \omega
 \end{aligned}$$

$$\therefore Y(\omega) = \frac{4}{\omega^2} \sin^2 \omega$$